ASSIGNMENT 4

Q2: Student Form Validator

Instructions: Save JavaScript-only solutions as `.js` files and run with Node.js. For form questions, save the full code as `.html` and open it in a browser. No frameworks are used.

Solution

```
<!DOCTYPE html>
<html lang="en"><head><meta charset="UTF-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Student Registration</title>
<style>
body{font-family:Arial;max-width:450px;margin:30px auto}
label{display:block;margin:12px 0}input{box-sizing:border-box;width:100%;padding:8px}
.valid{border:2px solid green}.invalid{border:2px solid red}.error{color:red}
</style></head><body>
<h1>Student Registration</h1>
<form id="form" novalidate>
<label>Name<input id="name"></label><small id="nameError" class="error"></small>
<label>Email<input id="email" type="email"></label><small id="emailError" class="error"></small>
<label>Phone<input id="phone" inputmode="numeric" maxlength="10"></label><small id="phoneError" class="error"></small>
<label>Password<input id="password" type="password"></label><small id="passwordError" class="error"></small>
<button>Register</button></form><p id="result"></p>
<script>
const rules={
 name:/^[A-Za-z]+$/,
 email:/^[^\s@]+@[^\s@]+\.[^\s@]+$/,
 phone:/^\d{10}$/,
 password:/^(?=.*[A-Z])(?=.*\d)(?=.*[^A-Za-z0-9]).+$/
};
const messages={
 name:"Use alphabets only.",email:"Enter a valid email address.",
 phone:"Phone must contain exactly 10 digits.",
 password:"Include an uppercase letter, a number and a special character."
};
function validate(id){
 const input=document.getElementById(id), valid=rules[id].test(input.value.trim());
 input.classList.toggle("valid",valid);input.classList.toggle("invalid",!valid);
 document.getElementById(id+"Error").textContent=valid?"":messages[id];return valid;
}
Object.keys(rules).forEach(id=>document.getElementById(id).addEventListener("input",()=>validate(id)));
document.getElementById("form").addEventListener("submit",e=>{
 e.preventDefault();const valid=Object.keys(rules).map(validate).every(Boolean);
 document.getElementById("result").textContent=valid?"Registration details are valid!":"Please correct the errors.";
});
</script></body></html>
```

Expected result / testing

Open the HTML file in a browser. Valid fields get green borders; invalid fields get red borders and error messages.